ASTANA IT UNIVERSITY

Polyclinic No. 5 Booking System

Telemedicine / digital health, Astana

Kira Kogay · Altynay Aitbayeva · Medet Azimkhan

Problem, users and what changed



Problem

Busy phone line or early queue. Paper schedule causes double bookings.



Main users

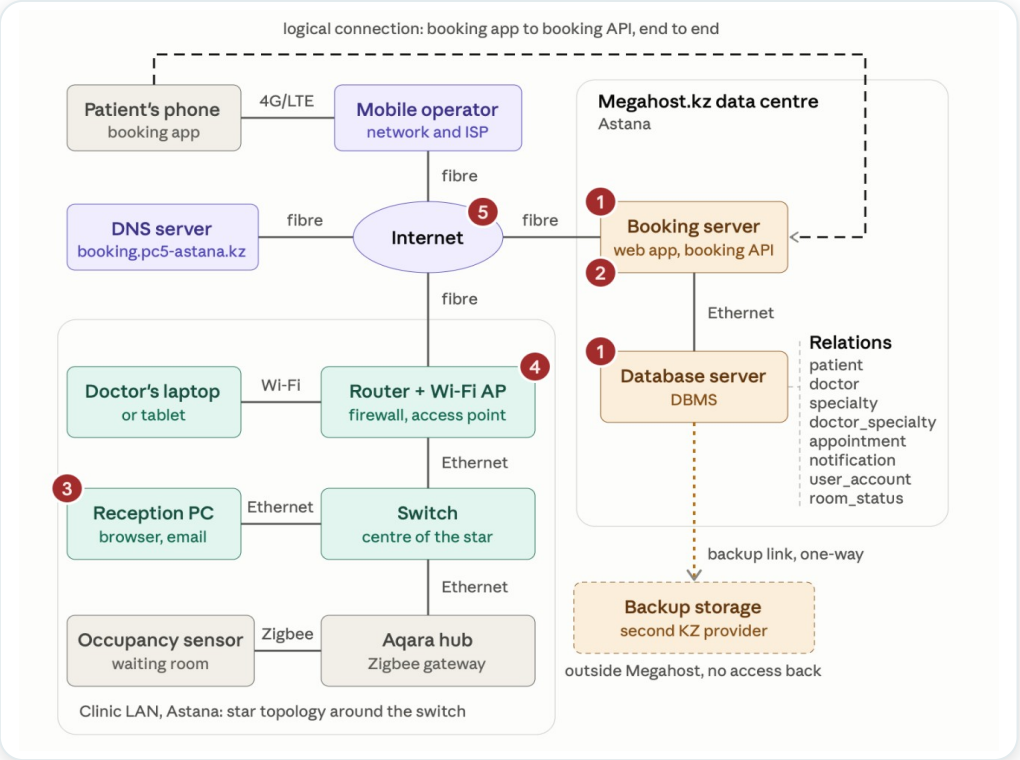
Patients, many elderly with weak eyesight.



After defense

Taken slot no longer a dead end: patient returns to the slot list.

System map



Phone to server over 4G and internet

Database + one-way backup

Aqara sensor via Zigbee hub

5 numbered threats

From URL to stored record

https://booking.pc5-astana.kz/doctors/12/slots

1. **DNS** phone gets the server IP
2. **TCP** SYN, SYN-ACK, ACK
3. **HTTPS** encrypted POST with the slot
4. **IP** packets routed to the data centre
5. **TCP** packets rebuilt in order
6. **SQL** appointment + notification saved
7. **HTTPS** reference number shown



HTTPS
protects logins on café Wi-Fi

Durability
keeps the record if the connection drops

Cloud and phone

Public cloud

- Megahost.kz, Astana
- Personal data stays in Kazakhstan
- Elasticity is manual: flu season, 8:00 rush

Phone + cloud

Issue: weak coverage in the clinic

Fix: slot stays on screen, Retry button, confirmed only after the server replies

One record: an appointment

Capture — Wk 8
Phone, JSON

Transmission — Wk 5
4G, HTTPS

Storage — Wk 6
PostgreSQL, Astana

Access — Wk 6
Row-level security

Protection — Wk 7
Roles, GRANT

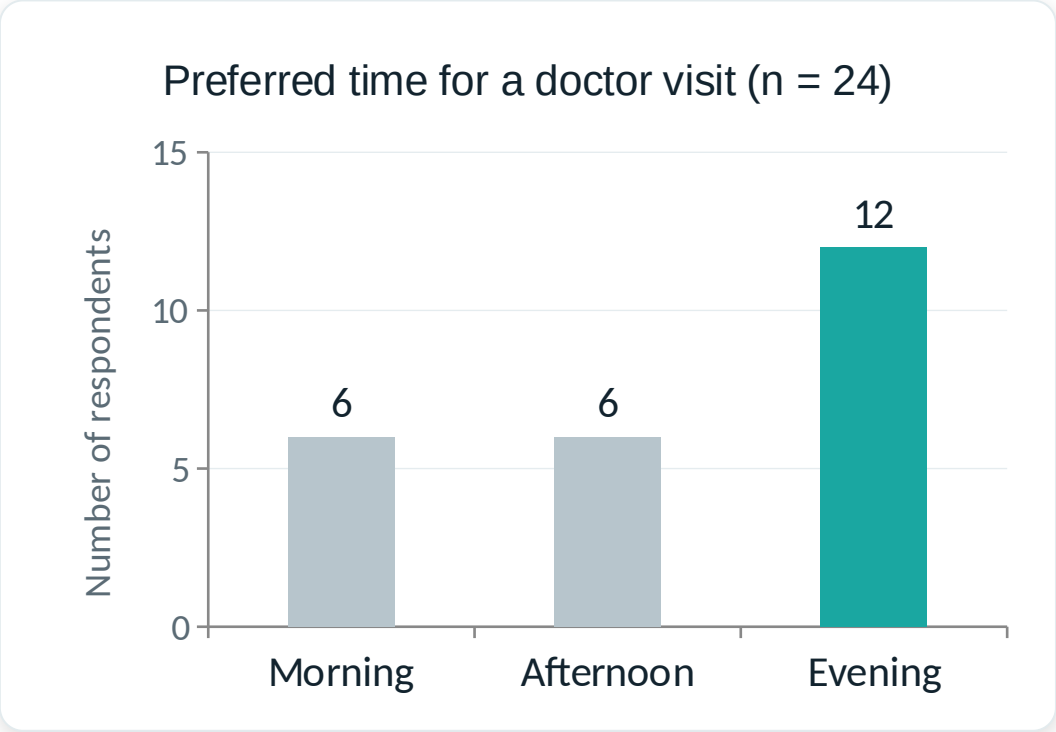
Backup — Wk 7
pgBackRest, daily

Analysis — Wk 9
Preference survey

Decision — Wk 9
Registry head

Riskiest stage: storage. It links identity to visit reasons.

Data analysis and decision



12 of 24

prefer an evening visit

Decision: open half of the slots in the evening, split the rest evenly. Keep some mornings for retired patients.